

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

```
Jser@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A2/IT314AB_Moreno-A (feature/data-ui)
$ git switch -c feature/student-list
Switched to a new branch 'feature/student-list'

Jser@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A2/IT314AB_Moreno-A (feature/student-list)
$ git branch
  feature/data-ui
* feature/student-list

Jser@DESKTOP-GL91V6M MINGW64 ~/Documents/Moreno_A2/IT314AB_Moreno-A (feature/student-list)
$
```

🚩 Flag 1 — Build Your Student Data

Screenshot:

```

Profile(
  name: "Kiyotaka Ayanokoji",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Reading, exercising",
  height: 5.9,
  isStudent: true,
  imagePath: "assets/ayanokoji.jfif",
  color: Colors.purple
),

Profile(
  name: "Suzune Horikita",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Reading, studying",
  height: 5.1,
  isStudent: true,
  imagePath: "assets/Suzune Horikita.jfif",
  color: Colors.red
),

```

```

Profile(
  name: "Kakeru Ryuen",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 17,
  hobby: "Fighting, strategizing",
  height: 5.9,
  isStudent: true,
  imagePath: "assets/Kakeru Ryuen 2.jfif",
  color: Colors.green
),

Profile(
  name: "Kikyo Kushida",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Socializing, making friends",
  height: 5.1,
  isStudent: true,
  imagePath: "assets/Kikyo Kushida.jfif",
  color: Colors.blue
),

```

```

Profile(
  name: "Airi Sakura",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Photography, singing",
  height: 5.0,
  isStudent: true,
  imagePath: "assets/Airi Sakura.jfif",
  color: Colors.orange
),
];

```

🚩 Flag 2 — The Copy-Paste Trap

Screenshot:

```

Profile(
  name: "Kiyotaka Ayanokoji",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Reading, exercising",
  height: 5.9,
  isStudent: true,
  imagePath: "assets/ayanokoji.jfif",
  color: Colors.purple
),

Profile(
  name: "Suzune Horikita",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Reading, studying",
  height: 5.1,
  isStudent: true,
  imagePath: "assets/Suzune Horikita.jfif",
  color: Colors.red
),

Profile(
  name: "Kakeru Ryuen",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 17,
  hobby: "Fighting, strategizing",
  height: 5.9,
  isStudent: true,
  imagePath: "assets/Kakeru Ryuen 2.jfif",
  color: Colors.green
),

Profile(
  name: "Kikyo Kushida",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Socializing, making friends",
  height: 5.1,
  isStudent: true,
  imagePath: "assets/Kikyo Kushida.jfif",
  color: Colors.blue
),

Profile(
  name: "Airi Sakura",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Photography, singing",
  height: 5.0,
  isStudent: true,
  imagePath: "assets/Airi Sakura.jfif",
  color: Colors.orange
),
];

```

🚩 Flag 3 —

Screenshot:

```

Profile(
  name: "Kiyotaka Ayanokoji",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Reading, exercising",
  height: 5.9,
  isStudent: true,
  imagePath: "assets/ayanokoji.jfif",
  color: Colors.purple
),

Profile(
  name: "Suzune Horikita",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Reading, studying",
  height: 5.1,
  isStudent: true,
  imagePath: "assets/Suzune Horikita.jfif",
  color: Colors.red
),

```

```

Profile(
  name: "Kakeru Ryuen",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 17,
  hobby: "Fighting, strategizing",
  height: 5.9,
  isStudent: true,
  imagePath: "assets/Kakeru Ryuen 2.jfif",
  color: Colors.green
),

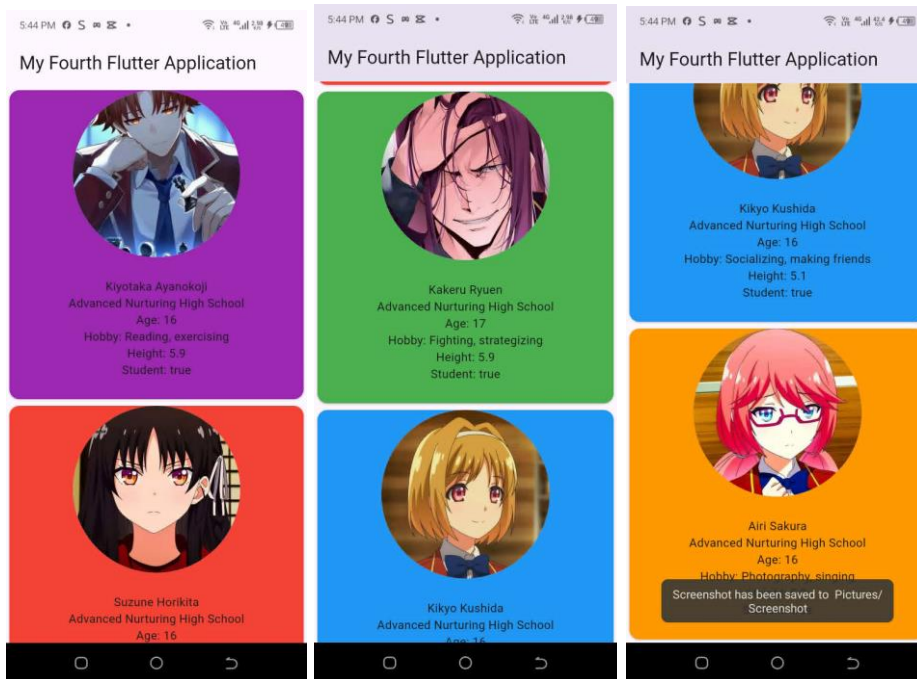
Profile(
  name: "Kikyo Kushida",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Socializing, making friends",
  height: 5.1,
  isStudent: true,
  imagePath: "assets/Kikyo Kushida.jfif",
  color: Colors.blue
),

```

```

Profile(
  name: "Airi Sakura",
  courseSection: "Advanced Nurturing High School",
  yearLevel: "1st Year",
  age: 16,
  hobby: "Photography, singing",
  height: 5.0,
  isStudent: true,
  imagePath: "assets/Airi Sakura.jfif",
  color: Colors.orange
),
];

```



🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

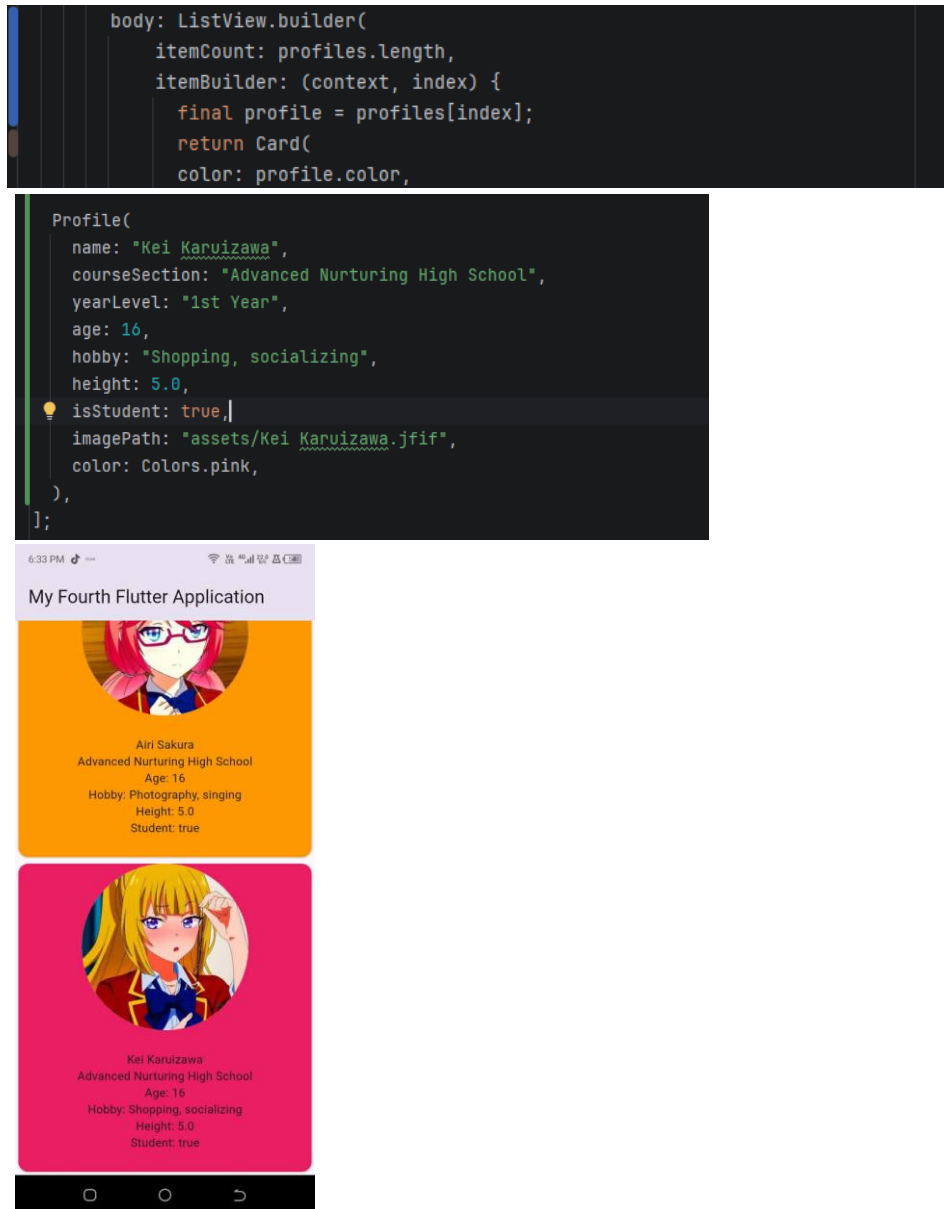
Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

Without creating another card manually, your application should display the new student automatically.

Screenshot:



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

- Student ID

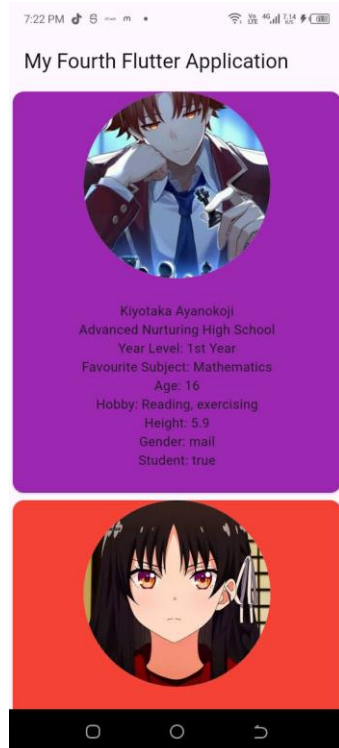
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

```
class Profile {  
    String? studentId;  
    String? name;  
    String? courseSection;  
    String? yearLevel;  
    String? favouriteSubject;  
    int? age;  
    String? hobby;  
    double? height;  
    String? gender;  
    bool isStudent;  
    String imagePath;  
    Color color;  
}
```

```
Profile({  
    required this.studentId,  
    required this.name,  
    required this.courseSection,  
    required this.yearLevel,  
    required this.favouriteSubject,  
    required this.age,  
    required this.hobby,  
    required this.height,  
    required this.gender,  
    required this.isStudent,  
    required this.imagePath,  
    required this.color,  
});  
}
```



🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

```
[]
```

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

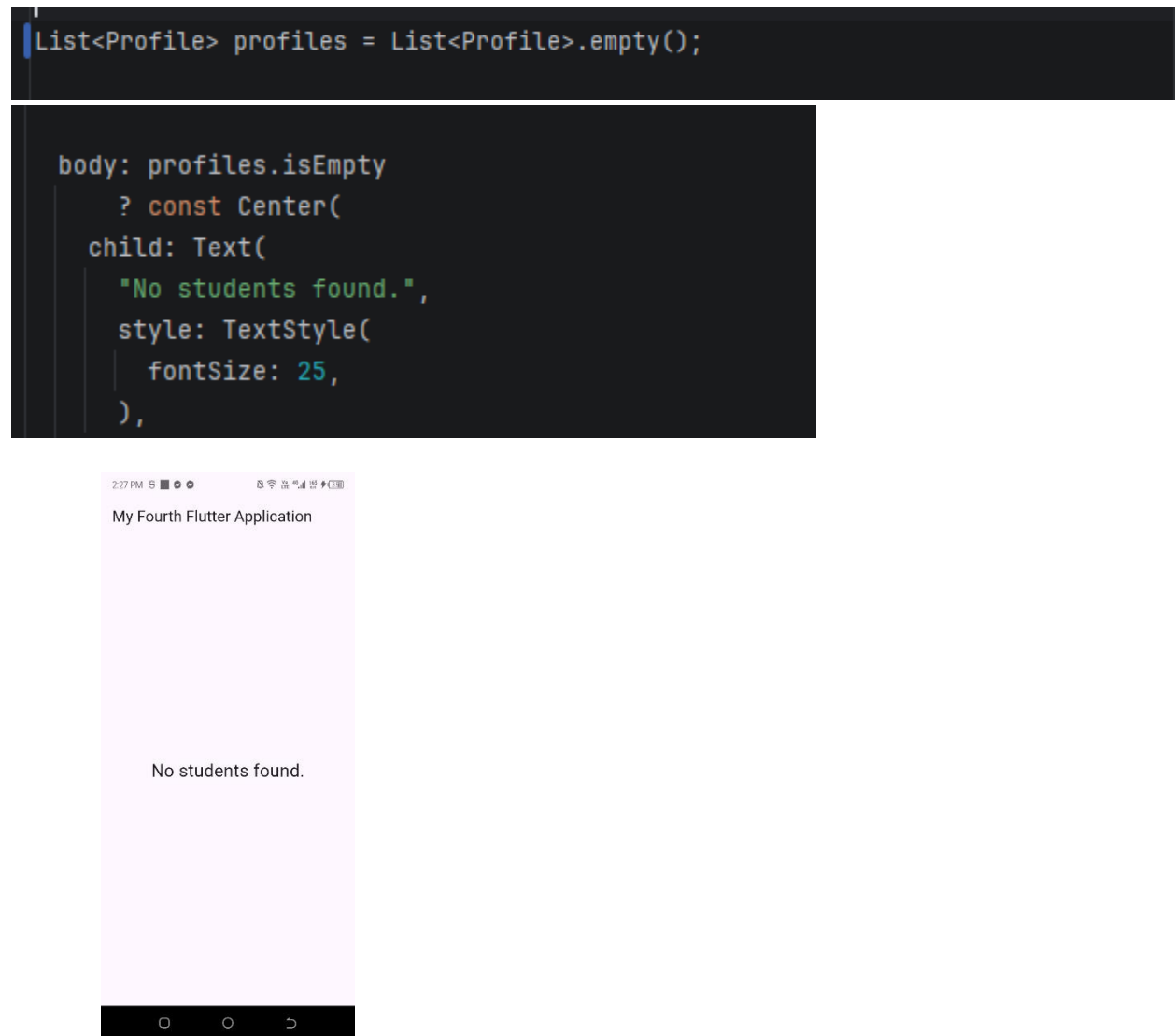
Examples:

- "No students found."
- "Student list is empty."

Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

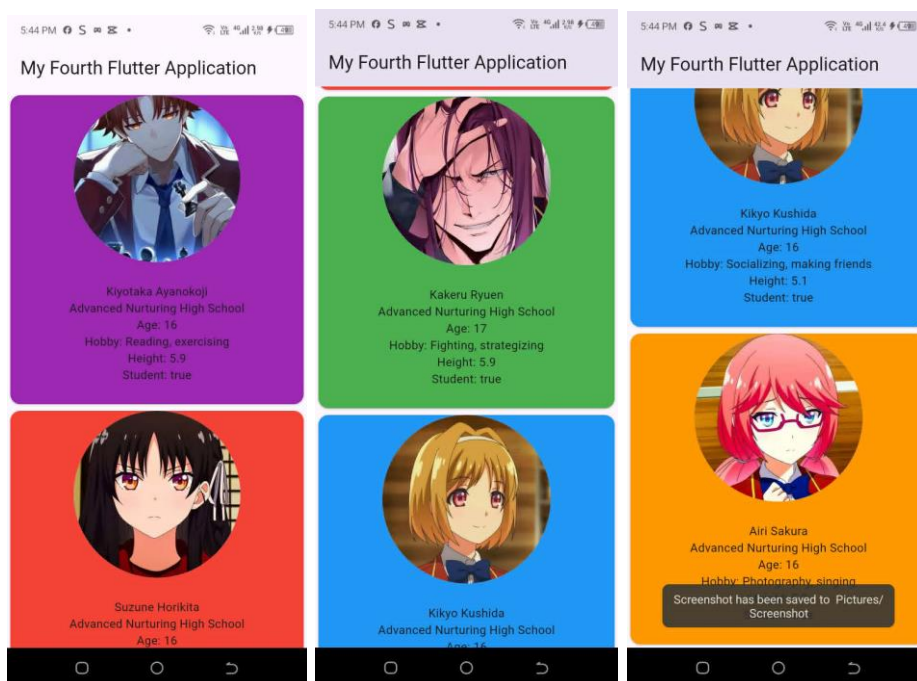


🚩 Flag 7 — Sort the Roll Call

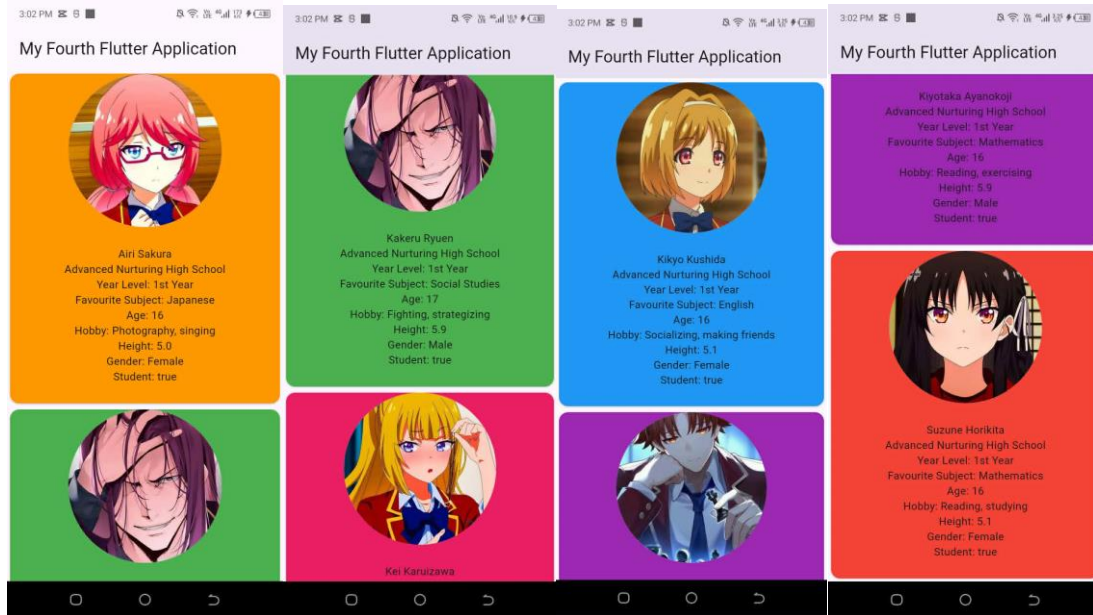
Screenshot:

```
void main() {
  profiles.sort((a, b){return a.name!.compareTo(b.name!);
});
```

BEFORE:



AFTER:



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User\Documents\Moreno_A2\IT314AB_Moreno-A>git branch
  feature/data-ul
+ feature/student-list

C:\Users\User\Documents\Moreno_A2\IT314AB_Moreno-A>git add .
warning: in the working copy of '.idea/workspace.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it

C:\Users\User\Documents\Moreno_A2\IT314AB_Moreno-A>git commit -m "feat: Activity Flag4"
[feature/student-list b4ef807] feat: Activity Flag4
17 files changed, 184 insertions(+), 2363 deletions(-)
 create mode 108644 my_first_flutter_app/assets/Alri Sakura.jfif
 create mode 108644 my_first_flutter_app/assets/Kakeru Ryuen 2.jfif
 create mode 108644 my_first_flutter_app/assets/Kakeru Ryuen.jfif

C:\Users\User\Documents\Moreno_A2\IT314AB_Moreno-A>git add .
warning: in the working copy of '.idea/workspace.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/linux/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_app/windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it

C:\Users\User\Documents\Moreno_A2\IT314AB_Moreno-A>git commit -m "feat: Activity Flag4"
[feature/student-list b4ef807] feat: Activity Flag4
17 files changed, 184 insertions(+), 2363 deletions(-)
 create mode 108644 my_first_flutter_app/assets/Alri Sakura.jfif
 create mode 108644 my_first_flutter_app/assets/Kakeru Ryuen 2.jfif
 create mode 108644 my_first_flutter_app/assets/Kakeru Ryuen.jfif
```